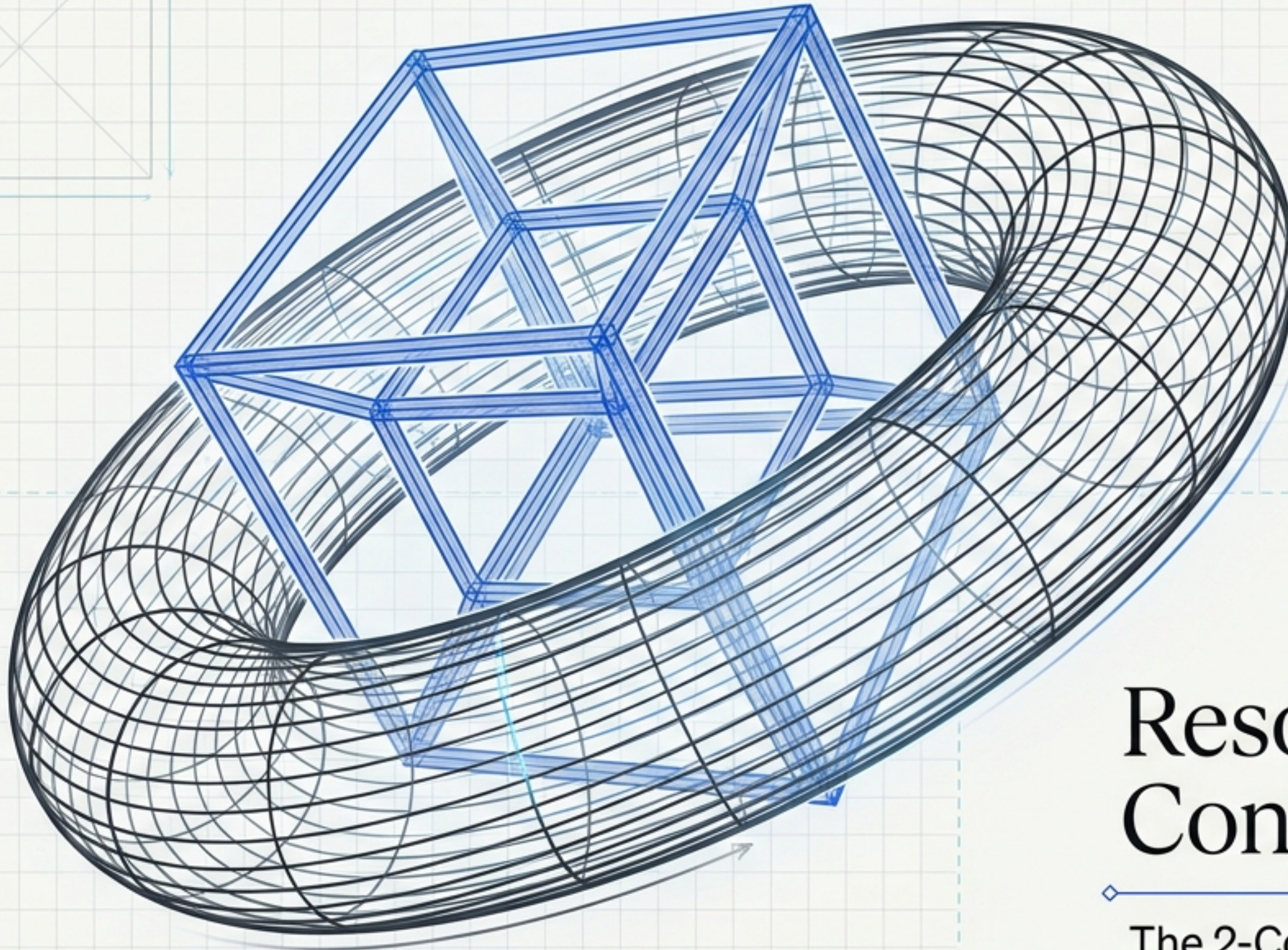




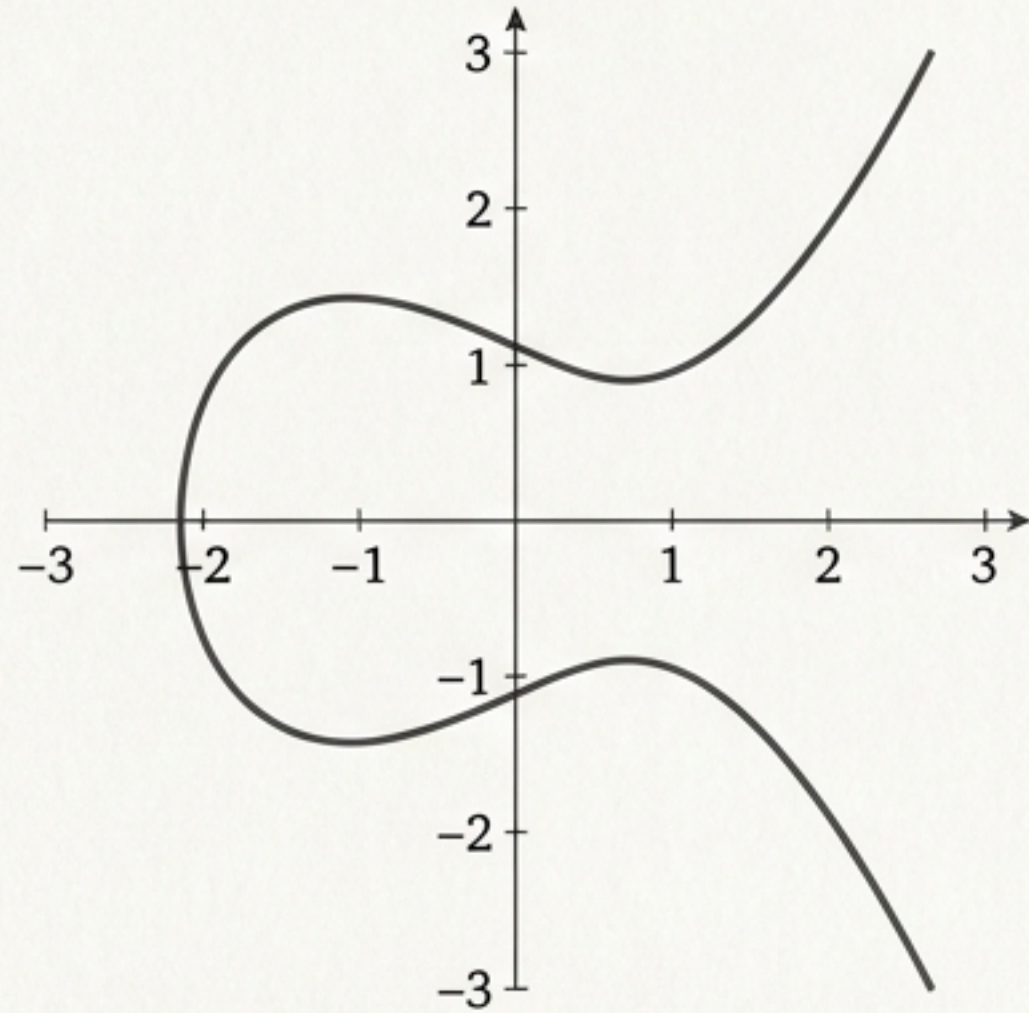
Resolving Goldfeld's Conjecture

The 2-Converse, Mean Analytic Rank, and the 2026 AI-Produced Proofs.

The Architecture of a Quadratic Twist

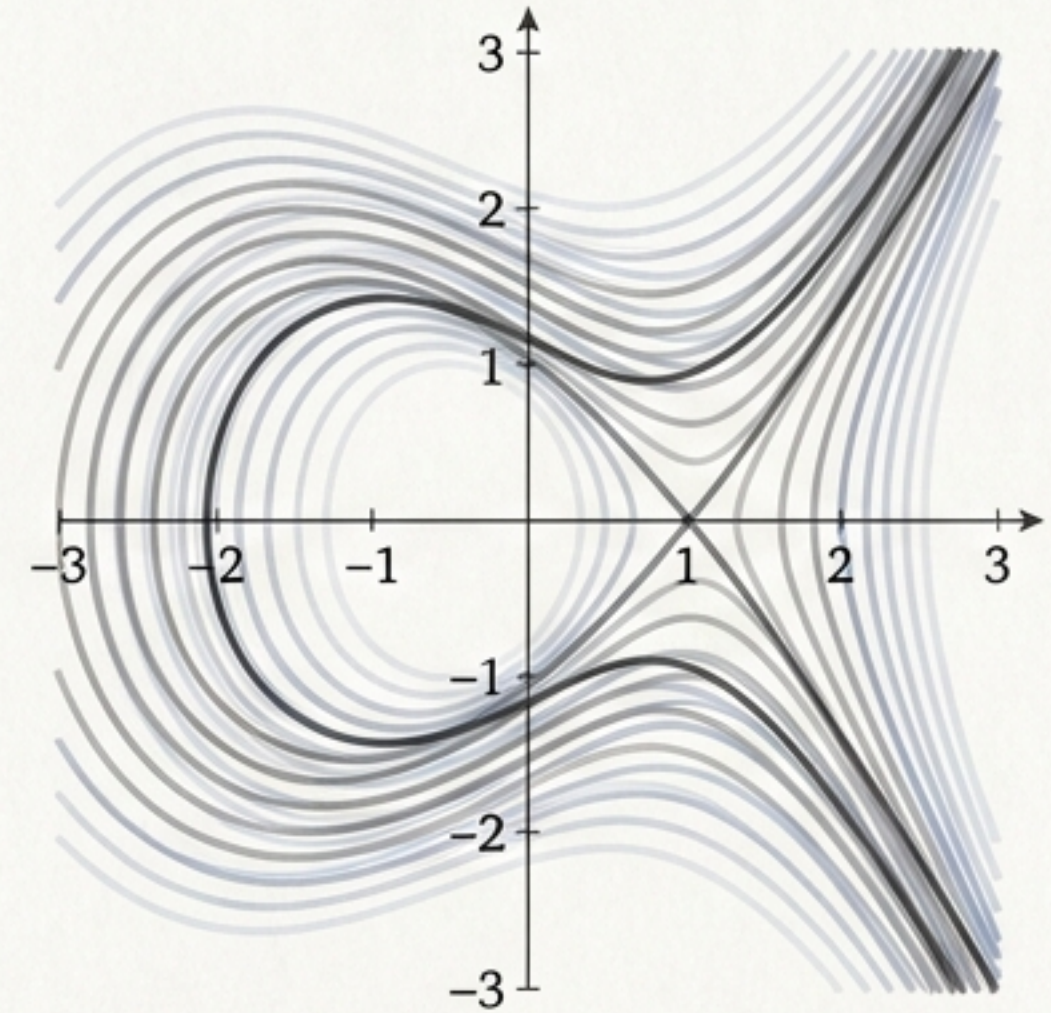
Base Curve E

$$y^2 = x^3 + ax + b$$



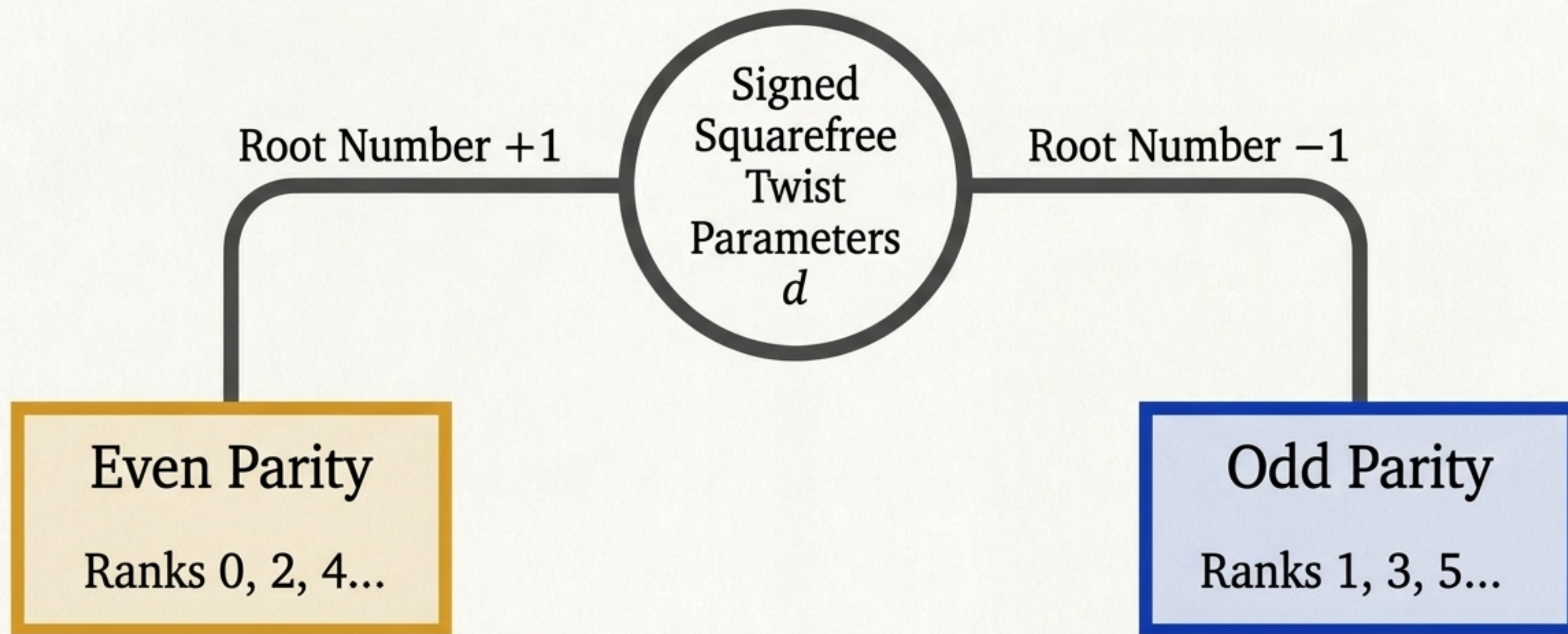
The Twist $E^{(d)}$

$$dy^2 = x^3 + ax + b$$



Analytic Rank $a(E)$ is the order of vanishing of the L-function $L(E, s)$ precisely at the critical point $s = 1$.

The Root Number Dictates a Perfect Parity Split

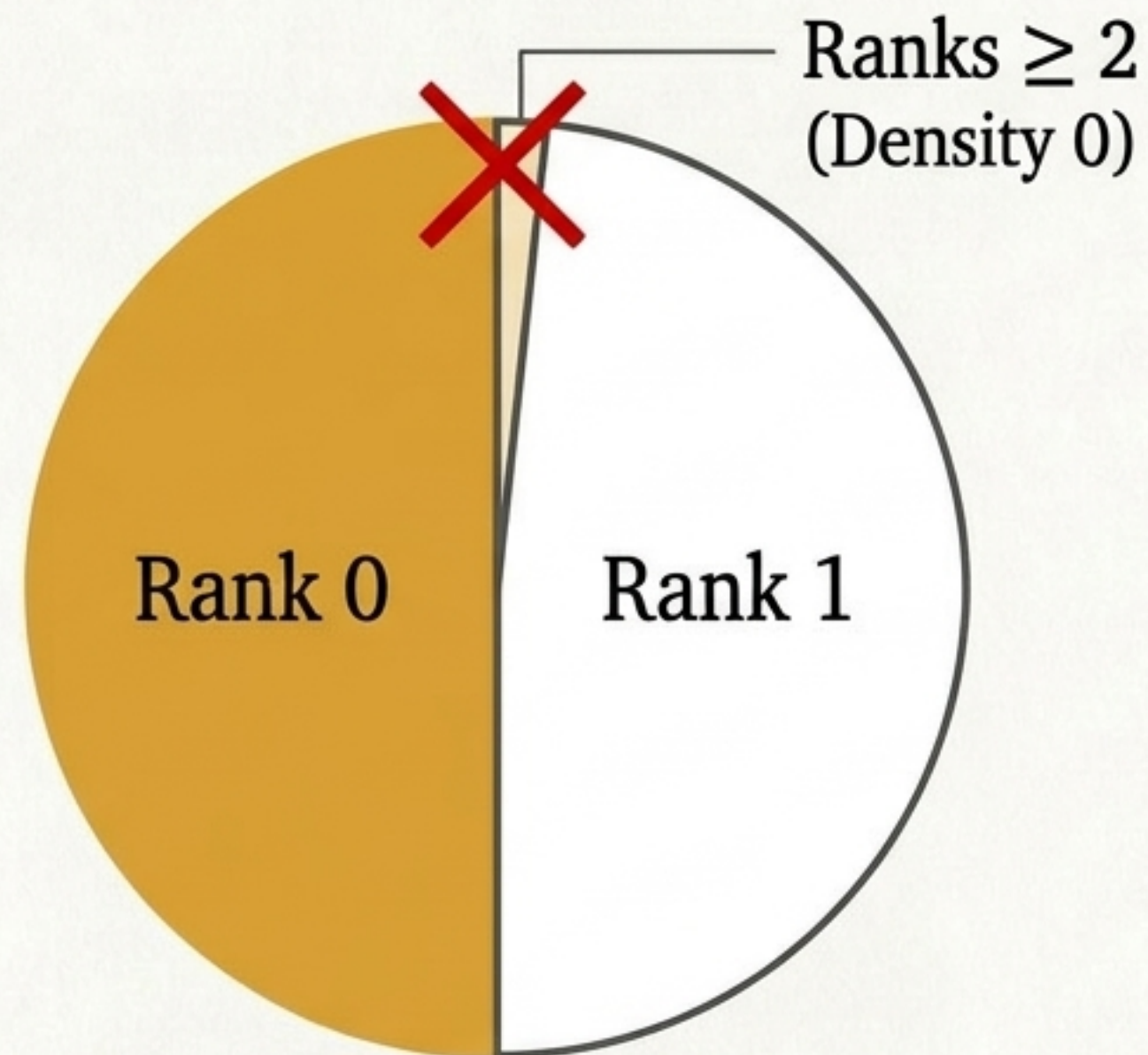


Each sign occurs for exactly half of all twists. The parity of the analytic rank is mathematically locked before any complex calculation begins.

Goldfeld's 1979 Conjecture



The average analytic rank of all twists converges to exactly $1/2$.



Predicted for families of twists parametrized by quadratic-field discriminants, ordered by absolute value $|d|$.

Four Decades of Partial Milestones

	Specific Families	General Curves	Selmer Bounds
Conditional on RH		Goldfeld bound 3.25 Heath-Brown bound $3/2$	
Unconditional	Vatsal for $X_0(19)$ Kriz-Li for rational 3-isogeny		Heath-Brown, Swinnerton-Dyer, Kane, Smith (2025)

We could prove properties for specific curves, bound averages assuming the Riemann Hypothesis, or track the Selmer group. But a universal, unconditional analytic proof eluded us.

The Roadblock: 2-Torsion in the Tate-Shafarevich Group

Poonen-Rains Random Model & Smith (2025)

Captures the 2-Selmer rank distribution (half 0, half 1)

The Selmer rank **over-counts** the actual rank for about 1/3 of twists because of **divisible 2-primary contributions** from the Tate-Shafarevich group (Sha).

[The Missing Link]

Analytic Rank



Breakthrough I: The Unrestricted 2-Converse

$$c_2(E) \in \{0, 1\} \Rightarrow a(E) = r(E) = c_2(E)$$

If the 2-infinity-Selmer
corank is 0 or 1...

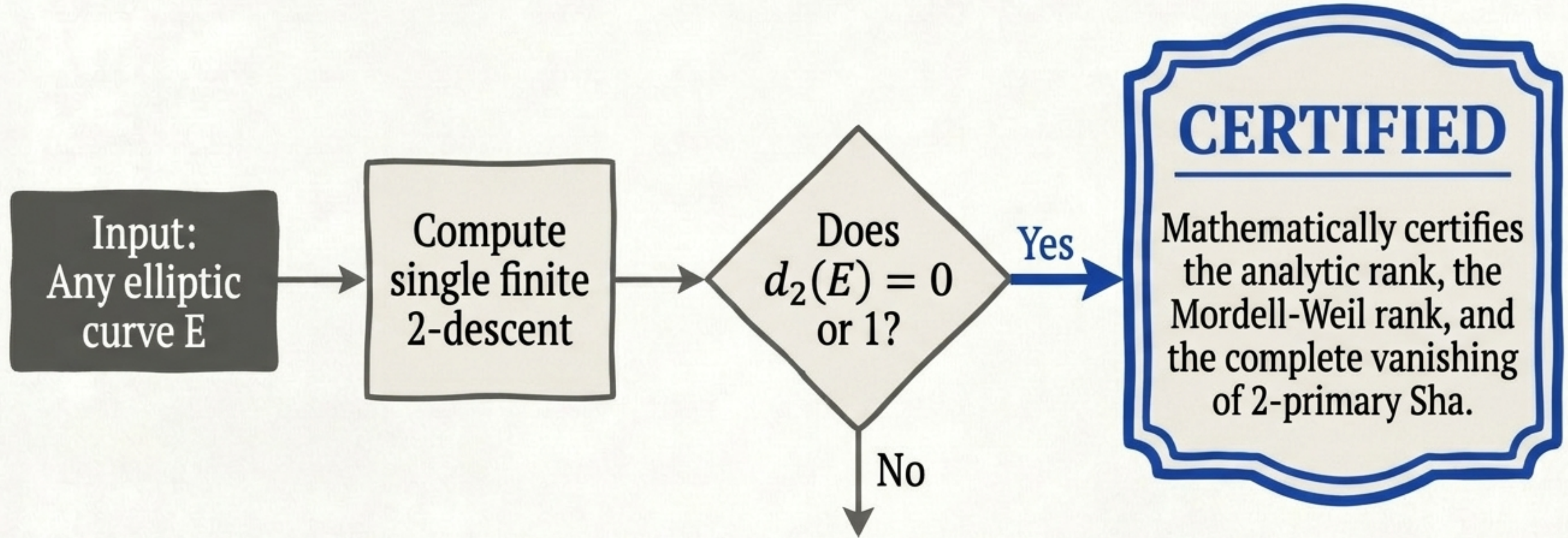
...it universally
forces...

...Analytic Rank to
equal Mordell-
Weil Rank...

...and guarantees a
finite Tate-Shafarevich
group (Sha).

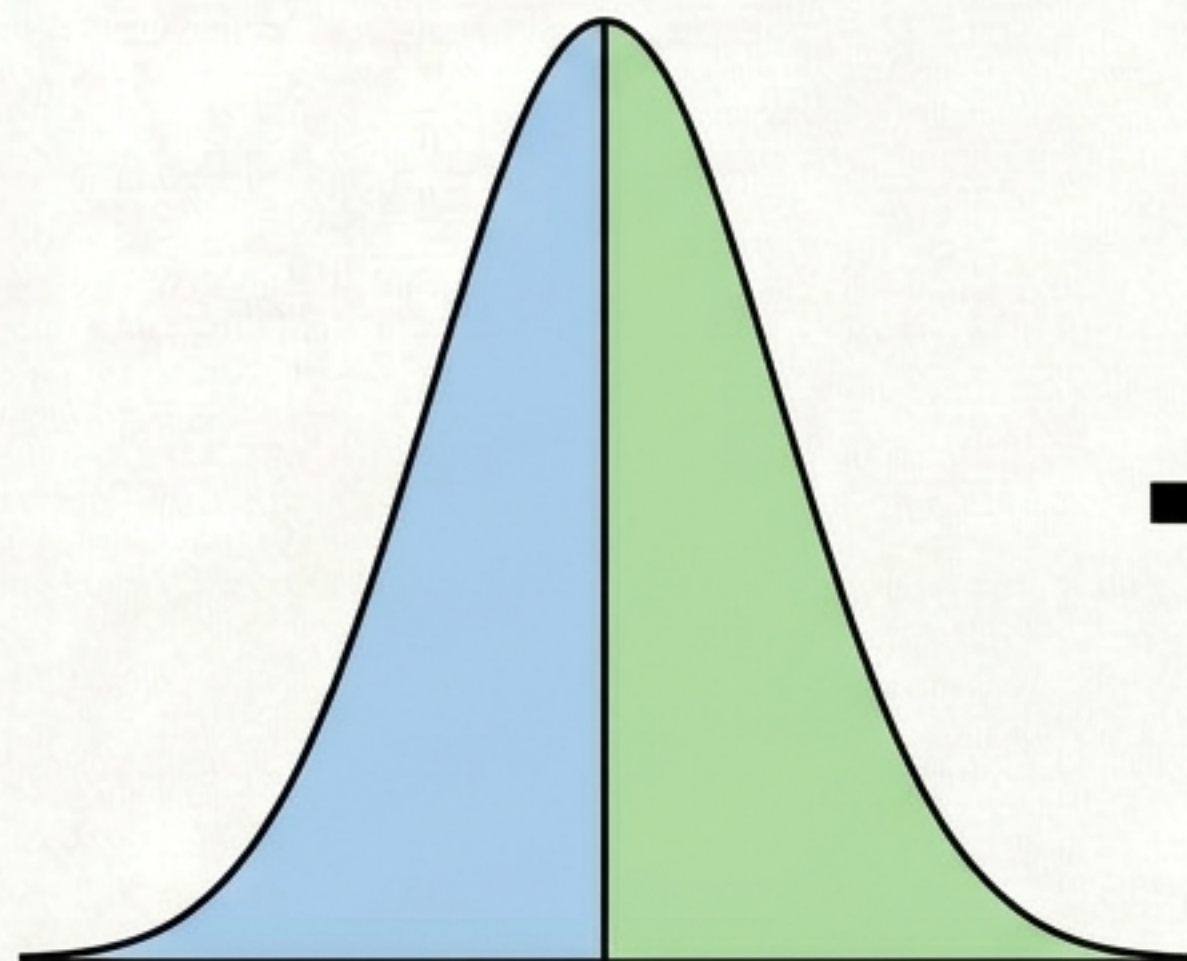
Theorem 1.1 bridges the gap. It applies to every elliptic curve over $\mathbb{Q}$
with no restrictions on reduction type, CM, or rational torsion.

The Pragmatic Consequence: Certifying Rank via Finite Descent



Corollary 1.3: Before this, finite descent could bound the rank, but could not certify the analytic rank or kill the Sha ambiguity unconditionally.

Breakthrough II: Proving Goldfeld's Densities



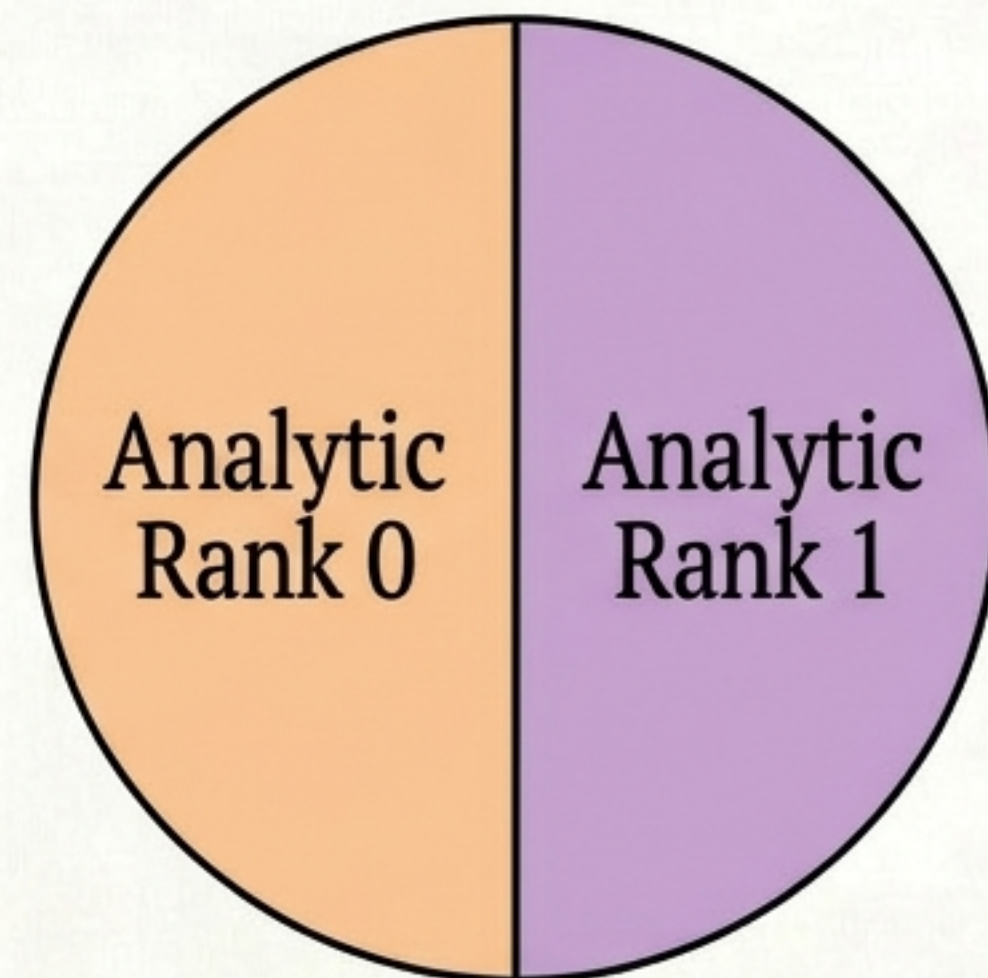
Smith (2025):
2-Selmer coranks

+



Theorem 1.1:
The 2-Converse

=



Theorem 1.2

Valid for every elliptic curve over $\mathbb{Q}$, counting signed squarefree twist parameters d ordered by absolute value $|d|$.

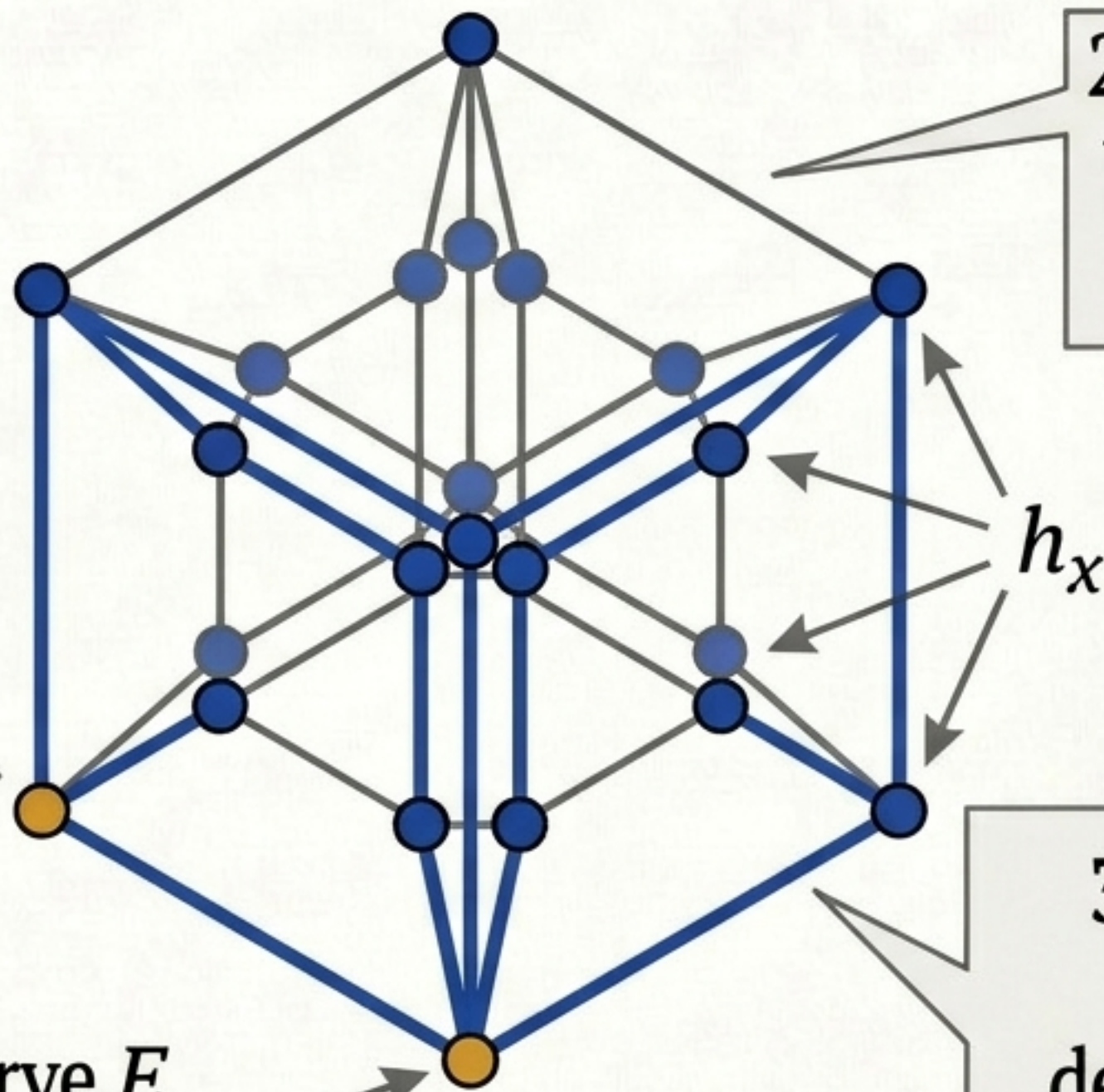
Proof Architecture: The Binary Cube of Twists

1 Place the target curve E at the zero vertex.

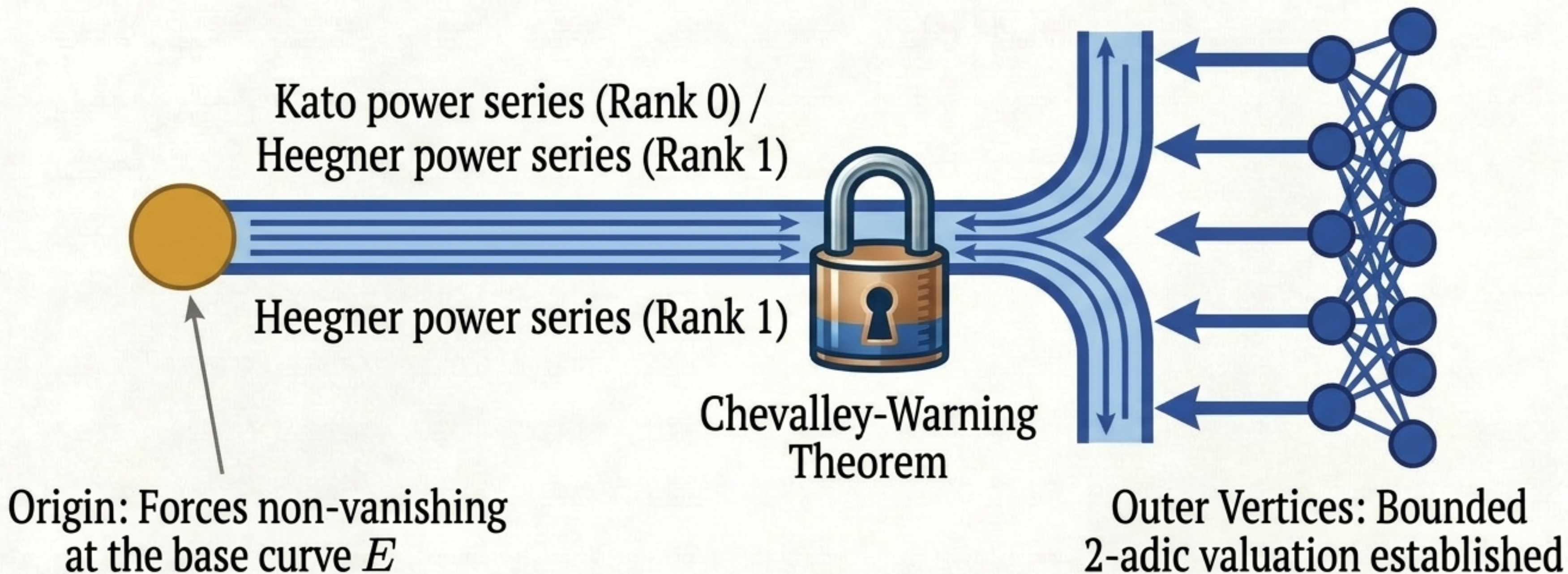
2 Construct a finite binary family of twists at all other vertices.

3. Rig modular-form coefficient tests to definitively succeed at non-zero vertex, forcing analytic non-vanishing.

Base Curve E
at origin
(0,0,...)

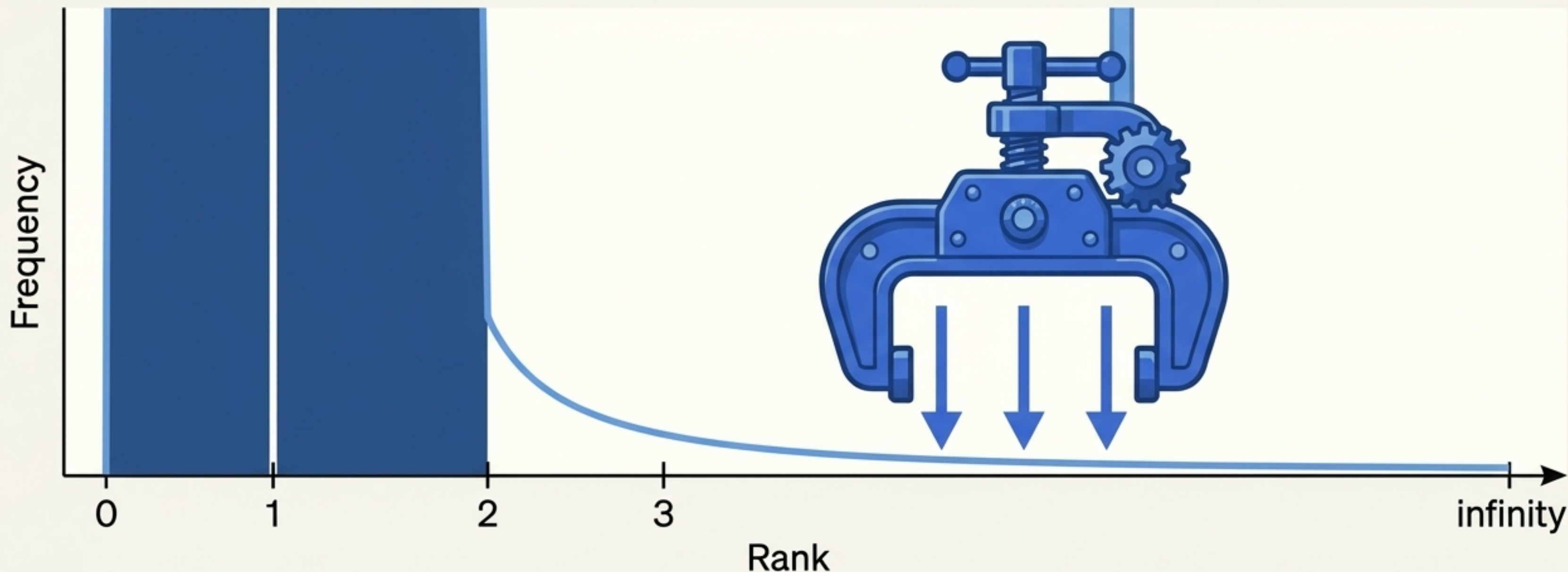


Proof Architecture: Valuation Transfer



This transfer clears denominators through a bounded complex, ensuring the congruence precision is independent of the number of new primes.

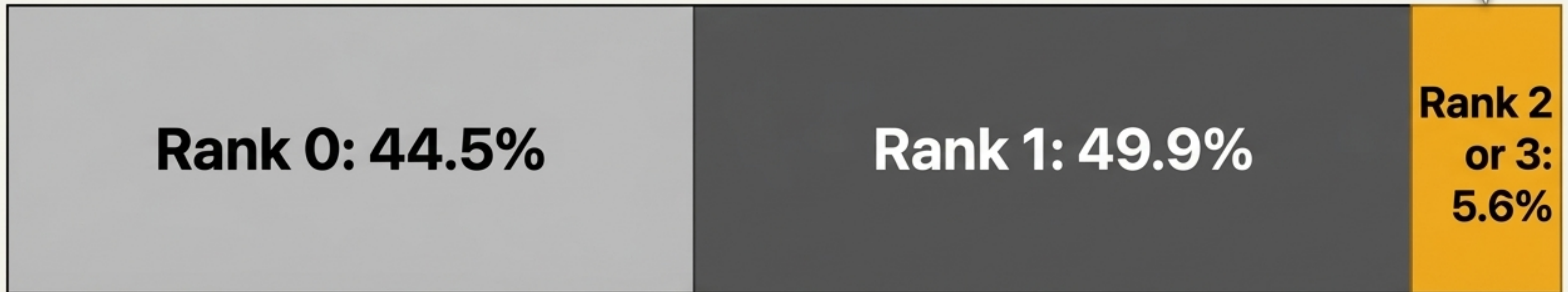
Breakthrough III: Bounding the Tail for Mean Rank



Knowing densities is not enough; rare twists of massive rank could skew the average. The companion paper enforces an analytic tail bound, proving that high-rank outliers are sparse enough that the mean converges exactly to $1/2$.

Empirical Reality: The Illusion of Excess Rank

Current Mean: 0.613



100% of the twists for curve 11a1 up to $|d| = 5000$

Theorem 1.1 and 1.2 describe an asymptotic limit as X approaches infinity. In bounded, real-world computation, rank ≥ 2 twists appear more frequently than their ultimate density of zero, demonstrating extremely slow convergence in practice.

The State of the Art & Lingerin Mysteries

The Result

Goldfeld's Conjecture is resolved for all rational curves.

Mean analytic rank is $1/2$, densities are exactly 50/50.

The Verification Gap

These preprints (Sep 2026) were generated by OpenAI.

The mathematical community now faces the challenge of human verification or waiting for a formalization in Lean.

The Open Frontier

Individual rank-2 twists remain an open mystery.

The 2-Converse master key only unlocks ranks 0 and 1; higher ranks remain uncertified by this mechanism.